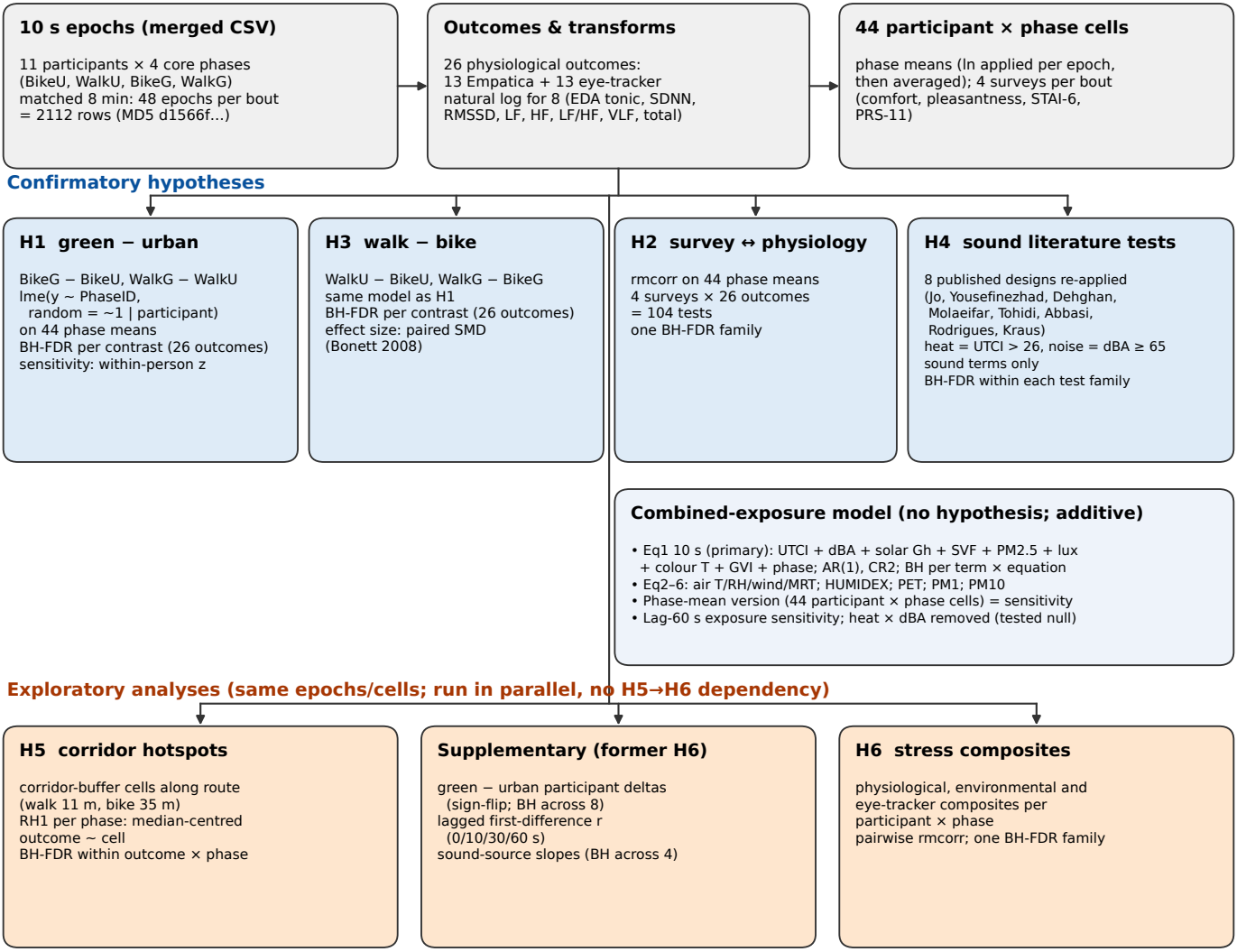
10 s epochs (merged CSV)
11 participants × 4 core phases
(BikeU, WalkU, BikeG, WalkG)
matched 8 min: 48 epochs per bout
= 2112 rows (MD5 d1566f...)
Outcomes & transforms
26 physiological outcomes:
13 Empatica + 13 eye-tracker
natural log for 8 (EDA tonic, SDNN,
RMSSD, LF, HF, LF/HF, VLF, total)
44 participant × phase cells
phase means (ln applied per epoch,
then averaged); 4 surveys per bout
(comfort, pleasantness, STAI-6,
PRS-11)
Confirmatory hypotheses
H1 green − urban
BikeG − BikeU, WalkG − WalkU
lme(y ~ PhaseID,
random = ~1 | participant)
on 44 phase means
BH-FDR per contrast (26 outcomes)
sensitivity: within-person z
H3 walk − bike
WalkU − BikeU, WalkG − BikeG
same model as H1
BH-FDR per contrast (26 outcomes)
effect size: paired SMD
(Bonett 2008)
H2 survey ↔ physiology
rmcorr on 44 phase means
4 surveys × 26 outcomes
= 104 tests
one BH-FDR family
H4 sound literature tests
8 published designs re-applied
(Jo, Yousefinezhad, Dehghan,
Molaeifar, Tohidi, Abbasi,
Rodrigues, Kraus)
heat = UTCI > 26, noise = dBA ≥ 65
sound terms only
BH-FDR within each test family
Combined-exposure model (no hypothesis; additive)
• Eq1 10 s (primary): UTCI + dBA + solar Gh + SVF + PM2.5 + lux
+ colour T + GVI + phase; AR(1), CR2; BH per term × equation
• Eq2–6: air T/RH/wind/MRT; HUMIDEX; PET; PM1; PM10
• Phase-mean version (44 participant × phase cells) = sensitivity
• Lag-60 s exposure sensitivity; heat × dBA removed (tested null)
Exploratory analyses (same epochs/cells; run in parallel, no H5→H6 dependency)
H5 corridor hotspots
corridor-buffer cells along route
(walk 11 m, bike 35 m)
RH1 per phase: median-centred
outcome ~ cell
BH-FDR within outcome × phase
Supplementary (former H6)
green − urban participant deltas
(sign-flip; BH across 8)
lagged first-difference r
(0/10/30/60 s)
sound-source slopes (BH across 4)
H6 stress composites
physiological, environmental and
eye-tracker composites per
participant × phase
pairwise rmcorr; one BH-FDR family